Subject: ACS 547, Module 4 Assignment - Sound Sources

Date: April 13, 2025 (Submitted)

Problem 1 - Simulation of Sources from Monopoles

The Matlab code for this problem is listed in Appendix 1.

Problem 1a

Figure 1 shows the spherical spreading from a monopole at 100 Hz, 1 kHz, and 10 kHz.

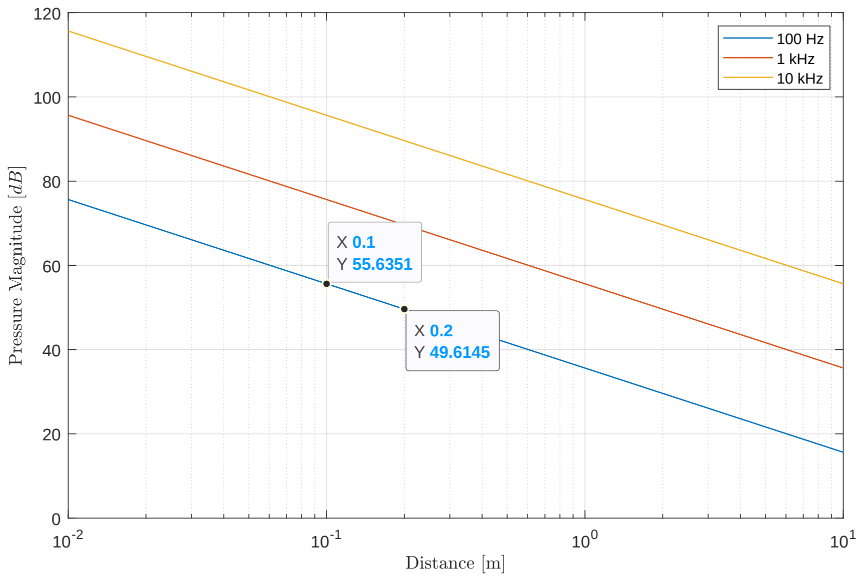

Figure 1: Monopole spherical spreading for 100 Hz, 1 kHz, and 10 kHz.

The pressure drop is proportional to $\frac{1}{\mathrm{r}}$, which is a 6 dB decay for a doubling of distance.

Problem 1b

Figure 2 shows Directivity patterns for a monopole, a dipole, a lateral quadrupole, and a longitudinal (linear) quadrupole at 100 Hz, 1 kHz, and 10 kHz.

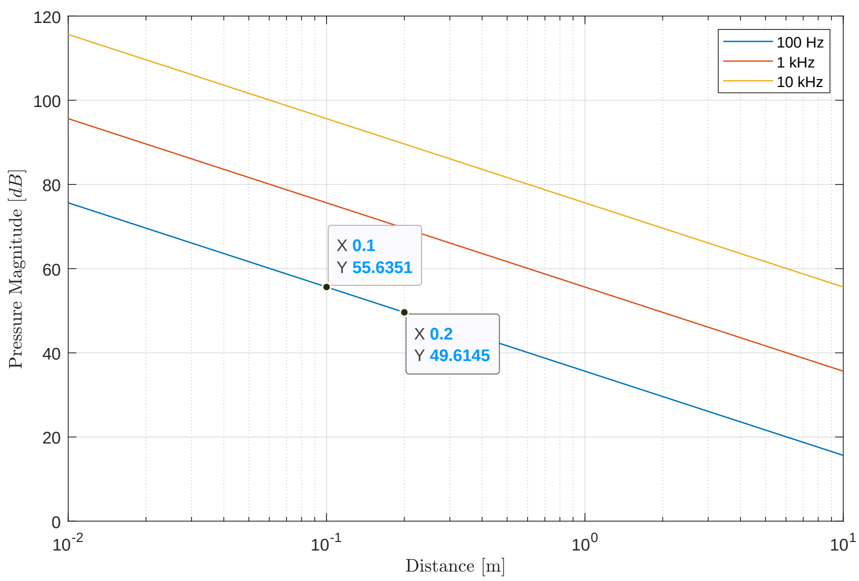

Figure 2: Directivity patterns for a monopole, a dipole, a lateral quadrupole, and a longitudinal (linear) quadrupole.

Problem 1c

Figure **??** shows the transmission loss of the system for a set of damping ratios ranging from 0.001 to 1.

Problem 1d

Figure **??** shows the force applied to the foundation.

No damping ratio makes the applied force less than 100 N across the 0 to 350 RPM operating range.

The best damping ratio is $\zeta = 0.25$. The force applied to the foundation still exceeds the 100 N target, but in only two small RPM regions: 91 to 109, and 331 to 350 RPM. In both of these regions, the maximum force is approximately 104 N.

Problem 1e

The viscous damping coefficient is 519.4 $\frac{\mathrm{kg}}{\mathrm{s}}$ based on a total mass of 110 kg.

Figure **??** shows the admittance.

The admittance at 300 RPM is 1e-5 $\frac{\mathrm{m}}{\mathrm{N}}$.

Problem 1f

Figure **??** shows the displacement of the washing machine.

The displacement of the washing machine at 300 RPM is approximately 4.9×10^{-3} m. The washing machine undergoes larger momentary displacements as it spins up to 300 RPM, in particular around 100 RPM.

The maximum displacement is 9.4×10^{-3} m, which is very large for prolonged operation.

Problem 2 - Washing Machine Mount Design (2 DOF)

The Matlab code for this problem is listed in Appendix 2.

Problem 2a

Figure **??** illustrates the two-stage vibration mount showing the washing machine and the “raft”.

Problem 2b

Table 1 lists the initial stiffness and damping coefficients for the system.

M1	110	kg
K1	100	$\frac{N}{m}$
C1	5	$\frac{kg}{s}$ [1]

M2	100	kg
K2	100	$\frac{N}{m}$
C2	5	$\frac{kg}{s}$ [2]

M3	0	kg
K3	1,000	$\frac{N}{m}$
C3	0	$\frac{kg}{s}$ [3]

Table 1: Initial values of stiffness and damping coefficients.

Damping was implemented using the damping vector[4].

[1] Element of damping vector.
[2] Element of damping vector.
[3] Element of damping vector.
[4] Not equivalent to using a complex-valued stiffness vector; the imaginary part is the damping for the respective spring

Problem 2c

Figure ?? shows the admittance of the washing machine and the raft.

The admittance of the washing machine is 9.31×10^{-6} $\frac{\mathrm{m}}{\mathrm{N}}$ at 300 RPM.

Problem 2d

Figure ?? show the displacement of the washing machine and the raft.

At 300 RPM, the washing machine displacement is 4.6×10^{-3} m, less than 5.0×10^{-3} m for the first-order system. The displacement of the raft is 4.7×10^{-5} m, which is 200 times less than the 1 DOF system.

Problem 2e

Figure ?? shows the force applied to the ground by the raft.

The force is is calculated by multiplying the displacement of the raft by the K2 spring constant.

The ground force at 300 RPM is 4.7×10^{-3} N. The force on the ground is always less than 100 N.

Problem 2f

The performance of this system was improved by *lowering the stiffness of the coupling spring,* k_3. Lowering k_3 decreases the separation of the two resonance peaks and shifts them to lower frequencies. This results in lower admittances for the washing machine and the raft at 300 RPM. However, due to the quadratic characteristic of the load force with RPM and the second-order roll-off of the admittance curve for the washing machine, optimization is restricted.

In Figure ??, displacement of the washing machine after the second resonance frequency "plateaus" with much less roll-off relative to the raft displacement. This limits optimization.

Problem 2g

Table 2 compares the admittance, displacement, and ground force for the initial and optimized systems.

	Admittance [$\frac{m}{N}$]	**Displacement [m]**	**Ground Force [N]**
Initial Design	9.31×10^{-3}	4.59×10^{-3}	4.7×10^{-3}
Optimized Design	9.22×10^{-6}	4.55×10^{-3}	4.62×10^{-5}

Table 2: Performance comparison of the initial and optimized designs.

It is apparent that the optimized design greatly reduces the admittance and the ground force. However, the change in displacement is small in comparison.

Problem 3 - Washing Machine Dynamic Vibration Absorber Design

The Matlab code for this problem is listed in Appendix 3.

Part 3a

Figure ?? illustrates the dynamic vibration absorber (DVA) system.

Table 1 lists the stiffness and damping coefficients for the system.

M1	15	kg
K1	0	$\frac{N}{m}$
C1	0	$\frac{kg}{s}$

M2	110	kg
K2	9,810	$\frac{N}{m}$
C2	1,962	$\frac{kg}{s}$ [1]

M3	0	kg
K3	1e5	$\frac{N}{m}$
C3	1e4	$\frac{kg}{s}$ [2]

Table 1: Stiffness and damping coefficients for the DVA system.

Damping was implemented by adding with an imaginary component to the respective spring stiffness[3].

[1] 0.2 fraction of respective spring stiffness.
[2] 0.1 fraction of respective spring stiffness.
[3] Not equivalent to using a damping vector in the argument to the given Matlab nDOF function script-file.

Part 3b

Figure ?? shows the admittance response of the original system and DVA system based on the equations presented in the tutorial lecture on DVA system analysis.

The washing machine admittance at 300 RPM is reduced from the blue curve resonance peak to a point between the two resonances on the red curve.

Part 3c - Admittance

Figure ?? shows the admittance response of the DVA system using the nDOF function script-file. The stiffness of the coupling spring, k_3, is set to place the null of the admittance response at the 300 RPM operating state of the washing machine. *The admittance at 300 RPM is approximately* $1.2 \times 10^{-6}\ \frac{\mathrm{m}}{\mathrm{N}}$[4]

Part 3d - Displacement

Figure ?? shows the displacement of the washing machine as a function of RPM. The ramp up to 300 RPM occurs quickly so that the large displacements are transient and the maximum speed is 350 RPM (the large displacement associated with the resonance peak at higher frequencies is not reached).

The displacement at 300 RPM is approximately 38.2×10^{-6} m.

Part 3e

Figure ?? shows the force applied to the floor by the washing machine.

The force applied to the floor at 300 RPM is approximately 3.75×10^{-3} N.

[4]This is theoretical and not realistic in practical terms, but the admittance is reduced.

Problem 4 - Washing Machine Abuse Testing

The Matlab code for this problem is listed in Appendix 4.

Part 4a

Equation 1 is the impulse response for a 1 DOF system.

$$x(t) = \frac{1}{m \cdot w_d} \cdot e^{(-w_o \cdot \zeta \cdot t)} \cdot \sin(w_d \cdot t) \tag{1}$$

where m is the mass of the washing machine, w_d is the damped natural frequency, w_o is the natural frequency, and ζ is the damping ratio.

Figure **??** shows plots of a 1 kg impulse force and the corresponding response of the washing machine.

Part 4b

A operational sequence for the washing machine created using a sinusoidal forcing function and the impulsive loading with the 5 kg brick. The sequence is as follows,

1. Washing machine is off.
2. Sinusoidal loading starts at 1 second.
3. 5 kg brick is inserted at 4 seconds.
4. Sinusoidal loading ends at 8 seconds.

Figure **??** shows the response sequence of the washing machine.

The maximum displacement of the brick response is 9.2×10^{-3} m.

The steady-state displacement is 5.4×10^{-3} m, which agrees with 4.9×10^{-3} m found in Problem 1f.

Part 4c

The washing machine takes about 2.3 seconds to return to a steady-state response.

Part 4d

Figure **??** shows the velocity response of the washing machine.

Figure **??** shows the acceleration response of the washing machine.

The maximum acceleration associated with the brick response is 500 $\frac{\mathrm{m}}{\mathrm{s}^2}$.

Summary

Table 1 compares the admittance, displacement, and ground force for the three systems considered here.

	Admittance [$\frac{\mathbf{m}}{\mathbf{N}}$]	**Displacement [m]**	**Ground Force [N]**
1-DOF	1×10^{-5}	4.9×10^{-3}	4.8×10^{1}
2-DOF Optimized	9.22×10^{-6}	4.6×10^{-3}	4.6×10^{-5}
DVA	1.2×10^{-6}	3.8×10^{-6}	3.8×10^{-5}

Table 1: Performance comparison of the initial and optimized designs.

1 Appendix - Matlab Code for Problem 1

2 Appendix - Matlab Code for Problem 2

3 Appendix - Matlab Code for Problem 3

4 Appendix - Matlab Code for Problem 4